THE UNIVERSITY OF SHEFFIELD

Introduction to Aerodynamic CFD

A rocket fin, from STEP file to lift coefficient

Student handout · ANSYS 2023 R2 / 2024 R1 · a three-hour workshop

What this document is

A complete, click-by-click record of the workshop: every number you type, and the reason it is that number rather than another.

What you need

ANSYS Workbench with Fluent and Discovery, the file `fin.step` from the course folder, and roughly 6 GB of free disk.

Version 1.0. Prepared for undergraduate teaching.

How to use this handout

The workshop moves through four stages — geometry, mesh, solution, post-processing. Each stage has the same shape:

- **Bold sans** is a button, tab, panel or dropdown you can click.
- **Purple bold** is a value you type or select.
- **File ▸ Preferences** is a menu path: click **File**, then **Preferences**.
- `monospace` is a file name or a name in a tree.

Element	What it does
Why	the theory that makes the next click defensible.
Do this	the numbered sequence of clicks and the exact values.
Checkpoint	what the screen should look like when the step worked.
If it goes wrong	the failures that actually happen, and the fix.

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1 The case: a fin, and the four numbers that define it

1.1 Geometry

The course folder contains `fin.step`: a model rocket fin with a swept leading edge, a trapezoidal planform and a sharp trailing edge.

> **Why the Reynolds number comes first**
> Every decision downstream — the size of the first cell at the wall, the density of the wake — depends on the Reynolds number. It is fixed by the chord, the speed and the air, and nothing later in the workflow can repair choosing it carelessly.

1.1 Set the operating point

Run the case at **U = 40 m/s**, sea level. With $c = 250\,\text{mm}$ this gives $Re \approx 5.1 \times 10^5$ — fully turbulent over most of the fin.

Values to enter

- Velocity magnitude: **40 m/s**
- Density (constant): **1.225 kg/m³**
- Viscosity: **1.7894e-05 kg/m s**

Checkpoint
The boundary-condition panel reads **velocity-inlet**, magnitude **40**, and the residual monitor starts near 10^{-1}.

If it goes wrong: the mesh will not fill
If the fill fails with a negative-volume error, the fluid body is not watertight: go back to §2.4, run the fault-isolating tool, and re-check the share topology before retrying.

Aside: the fin is thin, so the spanwise flow is weak; a symmetry treatment would be defensible but we model the full span for cleaner post-processing.

1.2 What the solver actually solves

For an incompressible, steady case Fluent solves the Reynolds-averaged momentum equations plus continuity; see the handout appendix for the discretisation settings and their defaults.

2 Closing

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